

Phase 4: Project Planning Phase

OptiCrop: Smart Agricultural Production Optimization

Role	Name
Team Lead	Anuhya Priya
Member 1	Dronadula Sai Chaitravi
Member 2	Srinu Buridi
Member 3	Andagundala Bhavana
Member 4	Naga Venkata Reddy Bethireddy

Project Roadmap

The project was executed over a 4-week timeline to ensure systematic progress from data ingestion to final deployment.

Timeline	Milestone & Tasks
Week 1	Environment configuration, dataset ingestion, and Exploratory Data Analysis (Distplots, Elbow Method).
Week 2	Data preprocessing (StandardScaler), model building (Logistic Regression, Decision Tree, Random Forest), evaluation, and model serialization (.pkl).
Week 3	Flask backend routing development and frontend web interface creation (HTML/CSS for Home, About, Predict pages).
Week 4	Integration of ML model with Flask, system testing (valid/invalid inputs), bug fixing, and documentation generation.